Testing AI Applications

AI applications should be tested for answer correctness, source accuracy, response time, and appropriate abstention. Correctness checks whether the answer matches the evidence. Source accuracy checks whether the cited document and page actually support the answer. Response time measures the delay between submitting a question and receiving a result.

A test set should include answerable questions, questions that require combining multiple passages, and questions whose answers are absent from the supplied documents.

Unsupported Questions and Prompt Injection

An assistant should decline to answer when the supplied documents do not provide sufficient evidence. A useful fallback is: I could not find this in the provided documents.

Documents may contain instructions that attempt to change the assistant's behavior, such as telling it to ignore previous instructions or reveal secrets. When documents are used as evidence, their contents should be treated as untrusted data, not as instructions that override the application's rules. The assistant should answer the user's question using relevant facts and ignore embedded behavioral commands.

Latency, Errors, and Usability

Latency can be measured with a monotonic clock around the retrieval and answer-generation operation. Error handling should explain invalid file types, unreadable PDFs, empty documents, and failures to retrieve relevant evidence.

A useful interface should show source references, make uncertainty visible, and provide concise answers. A local demonstration can use sample documents and a reproducible evaluation report; public hosting is optional when a working local demo satisfies the project requirements.